Pimpri-Chinchwad Education Trust's
Pimpri-Chinchwad College of Engineering
(An Autonomous Institute)
Affiliated to Savitribai Phule Pune University (SPPU)
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Assignment No 5

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Subject: Design and Analysis of Algorithms
Batch: A3
GitHub Link: https://github.com/SOHAM0007-CODER/DAA_ASSIGNMNET

1. Topic

Model SwiftCargo's warehouses, transit hubs and delivery points as a directed multistage graph. Find the fastest or lowest fuel-cost route through every stage. Support traffic and weather delays, road closures and multiple delivery requests.

2. Course Objective

- Understand the computational complexity of algorithms.
- Select a suitable algorithm design strategy for a practical problem.

3. Course Outcome

- Analyze a dynamic-programming route algorithm and its computational cost.
- Find valid stage-by-stage routes and replan when road conditions change.

Theory

A multistage graph groups cities into ordered stages. Each directed road goes from stage s to stage s+1. Therefore, a delivery passes through every stage and the graph has no cycle. Dynamic programming solves later stages first and uses those answers to find the best route from earlier stages.

Fastest mode uses base travel time + traffic delay + weather delay. Cheapest mode uses estimated fuel litres x fuel price. Only one objective is minimized in each request. Closed roads are ignored; fuel cost excludes tolls and other expenses.

Working of the Algorithm

- 1. Read city stages and road conditions; print the road table on request.
- 2. Solve backwards and print best remaining values for each city.
- 3. Display the optimal path, selected road IDs and per-leg delays/fuel/cost.
- 4. Process batches, close roads manually and reroute or report unreachable.

Complexity Analysis

Each city and road is processed a constant number of times. With V cities and E roads, one request takes O(V + E). A batch of Q requests takes O(Q(V + E)). Road updates by known ID take O(1).

Time Complexity

Best Case: O(V + E) per request.

Average Case: O(V + E) per request.

Worst Case: O(V + E) per request; the complete stage scan is performed.

Space Complexity

O(V + E) for adjacency lists, stage buckets and route arrays. This avoids a V-by-V adjacency matrix.

Algorithm - Steps

- Step 1: Read the city stages, fuel price, roads and requested endpoints.
- Step 2: Choose fastest or cheapest mode and solve backwards through the stages.
- Step 3: Print the route, road IDs, travel time, fuel use and fuel cost.

Pseudocode

```
best[destination] = 0; all other best values = infinity
for stage from last-1 down to 1:
    for each city u in stage:
        for each open road (u,v):
            if weight(u,v) + best[v] < best[u]:
                best[u] = weight(u,v) + best[v]
                nextRoad[u] = (u,v)
follow nextRoad from the source to print the route
```

Flow of Execution

START -> Read network -> Choose objective -> Backward DP -> Print route -> Update or process another batch -> END.

6. Example

Stages are [1,2,2,3], fuel price is 100 INR/litre. The fastest route 1 -> 2 -> 4 takes 12 minutes including delays and costs 400 INR. The cheapest route 1 -> 3 -> 4 takes 16 minutes and costs 200 INR. Closing road 1 reroutes the fastest request via city 3; closing road 2 as well makes delivery impossible.

Conclusion

Dynamic programming finds an optimal route for the selected objective in a staged network. Adjacency lists support efficient processing, while manual road updates and batch requests allow repeated planning with current conditions.

Assignment 5 | SAMPLE INPUT (input.txt)

```
4 3
1 2 2 3
100
4
1 2 5 1 0 2 1
1 3 8 0 0 1 1
2 4 5 0 1 2 1
3 4 8 0 0 1 1
3
1
2
1 4 1
1 4 2
2
1
5 1 0 2 0
1
1
1 4 1
2
2
8 0 0 1 0
1
1
1 4 1
0
```

Assignment 5 | FULL PROGRAM OUTPUT

```
ASSIGNMENT 5: SWIFTCARGO - MULTISTAGE ROUTING
Enter number of cities and stages:
Enter stage of each city:
Enter fuel price (INR per litre):
Enter number of roads:
Enter: from to baseTime trafficDelay weatherDelay fuel open(1/0)
Road 1:
Road 2:
Road 3:
Road 4:

1. Delivery batch  2. Update road  3. Show roads  0. Exit

ROAD TABLE
ID | From | To | Base | Traffic | Weather | Fuel L | Status
1 | 1 | 2 | 5.00 | 1.00 | 0.00 | 2.00 | OPEN
2 | 1 | 3 | 8.00 | 0.00 | 0.00 | 1.00 | OPEN
3 | 2 | 4 | 5.00 | 0.00 | 1.00 | 2.00 | OPEN
4 | 3 | 4 | 8.00 | 0.00 | 0.00 | 1.00 | OPEN

1. Delivery batch  2. Update road  3. Show roads  0. Exit
Enter number of requests:
Enter source destination mode(1=fastest, 2=cheapest):

DELIVERY REQUEST 1 OF 2 | 1 -> 4
Fastest delivery

DP TABLE: best remaining time (minutes) to city 4
City | Stage | Best remaining value | Next road
1 | 1 | 12.00 | 1
2 | 2 | 6.00 | 3
3 | 2 | 8.00 | 4
4 | 3 | 0.00 | -
Optimal route: 1 -> 2 -> 4
Road IDs: 1 3
LEG DETAILS
Road | From -> To | Base min | Traffic min | Weather min | Effective min | Fuel L | Cost INR
1 | 1 -> 2 | 5.00 | 1.00 | 0.00 | 6.00 | 2.00 | 200.00
3 | 2 -> 4 | 5.00 | 0.00 | 1.00 | 6.00 | 2.00 | 200.00
Stages covered: 3 / 3 | All route roads: OPEN

Total time: 12.00 minutes
Total fuel: 4.00 litres
Total fuel cost: 400.00 INR
```

```
Enter source destination mode(1=fastest, 2=cheapest):

DELIVERY REQUEST 2 OF 2 | 1 -> 4
Cheapest delivery

DP TABLE: best remaining fuel cost (INR) to city 4
City | Stage | Best remaining value | Next road
1 | 1 | 200.00 | 2
2 | 2 | 200.00 | 3
3 | 2 | 100.00 | 4
4 | 3 | 0.00 | -
Optimal route: 1 -> 3 -> 4
Road IDs: 2 4
LEG DETAILS
Road | From -> To | Base min | Traffic min | Weather min | Effective min | Fuel L | Cost INR
2 | 1 -> 3 | 8.00 | 0.00 | 0.00 | 8.00 | 1.00 | 100.00
4 | 3 -> 4 | 8.00 | 0.00 | 0.00 | 8.00 | 1.00 | 100.00
Stages covered: 3 / 3 | All route roads: OPEN

Total time: 16.00 minutes
Total fuel: 2.00 litres
Total fuel cost: 200.00 INR

1. Delivery batch  2. Update road  3. Show roads  0. Exit
Enter road ID:
Enter baseTime trafficDelay weatherDelay fuel open(1/0):
Road 1 updated: CLOSED. Future requests use these values.

1. Delivery batch  2. Update road  3. Show roads  0. Exit
Enter number of requests:
Enter source destination mode(1=fastest, 2=cheapest):

DELIVERY REQUEST 1 OF 1 | 1 -> 4
Fastest delivery

DP TABLE: best remaining time (minutes) to city 4
City | Stage | Best remaining value | Next road
1 | 1 | 16.00 | 2
2 | 2 | 6.00 | 3
3 | 2 | 8.00 | 4
4 | 3 | 0.00 | -
Optimal route: 1 -> 3 -> 4
Road IDs: 2 4
LEG DETAILS
```

Assignment 5 | FULL PROGRAM OUTPUT (continued)

```
Road | From -> To | Base min | Traffic min | Weather min | Effective min | Fuel L | Cost INR
2 | 1 -> 3 | 8.00 | 0.00 | 0.00 | 8.00 | 1.00 | 100.00
4 | 3 -> 4 | 8.00 | 0.00 | 0.00 | 8.00 | 1.00 | 100.00
Stages covered: 3 / 3 | All route roads: OPEN

Total time: 16.00 minutes
Total fuel: 2.00 litres
Total fuel cost: 200.00 INR

1. Delivery batch  2. Update road  3. Show roads  0. Exit
Enter road ID:
Enter baseTime trafficDelay weatherDelay fuel open(1/0):
Road 2 updated: CLOSED. Future requests use these values.

1. Delivery batch  2. Update road  3. Show roads  0. Exit
Enter number of requests:
Enter source destination mode(1=fastest, 2=cheapest):

DELIVERY REQUEST 1 OF 1 | 1 -> 4
Fastest delivery

DP TABLE: best remaining time (minutes) to city 4
City | Stage | Best remaining value | Next road
1 | 1 | Unreachable | -
2 | 2 | 6.00 | 3
3 | 2 | 8.00 | 4
4 | 3 | 0.00 | -
No delivery route exists.

1. Delivery batch  2. Update road  3. Show roads  0. Exit
Time per request: O(V+E) | Network storage: O(V+E)
```